

logue CCD camera along with a video decoder

3. A state-of-the-art quadrupole mass spectrometer with a mass resolution of 0.5 amu. It is sensitive to partial pressure of the order of 10-15 torr for measuring the constituents of tenuous lunar atmosphere during descent

In addition to the instruments, the separation system, the de-boost spin and de-spin motors, it comprises the avionics system and thermal control system. The avionics system supports the payloads and provides communication link between the moon impact probe and the main orbiter, from separation to impact, and provides a database useful for future soft landing. The mission envisages collection of all the instrument data during descent and transmission to the main orbiter, which, in turn, will transmit it to the ground station during visible phases.

Mission objectives

1. To harness the science payloads, lunar craft and the launch vehicle with suitable ground support systems in-

cluding deep space station

2. To realise the integration and testing, launch into the lunar polar orbit of about 100 km, in-orbit operation of experiments, communication/telecommand, telemetry data reception, quick-look analysis of data and archival for scientific utilisation by an identified group of scientists

Scientific objectives

The Chandrayaan-1 mission is aimed at high-resolution remote sensing of the moon in visible, near-infrared, low-energy X-ray and high-energy X-ray regions. Specifically, the objectives are:

1. To prepare a three-dimensional atlas (with a high spatial and altitude resolution of 5-10 metres) of both near and far sides of the moon

2. To conduct chemical and mineralogical mapping of the entire lunar surface for distribution of elements such as magnesium, aluminium, silicon, calcium, iron and titanium with a spatial resolution of about 25 km and high-atomic-number elements such as radon, uranium and thorium with a

spatial resolution of about 20 km

Simultaneous photo geological and chemical mapping will enable identification of different geological units, which will test the early evolutionary history of the moon and help in determining the nature and stratigraphy of the lunar crust.

Next in line: Chandrayaan-2

ISRO is also planning a second version of Chandrayaan named Chandrayaan-2. According to ISRO chairman G. Madhavan Nair, "The ISRO hopes to land a motorised rover on the Moon in 2012, as a part of its second Chandrayaan mission. The rover will be designed to move on wheels on the lunar surface, pick up samples of soil or rocks, do on-site chemical analysis and send the data to the mother-spacecraft Chandrayaan-2, which will be orbiting above. Chandrayaan-2 will transmit the data to the earth." ●

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